CAMPUS FOOD ORDERING APP

A PROJECT REPORT

Submitted by

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CHENNAI - 602105

BONAFIDE CERTIFICATE

Certified that this project report "**CAMPUS FOOD ORDERING APP**" is the bonafide work of "**ARAVINTHAA S (2116230701032)**" who carried out the project work (CS23621-Mobile Application Development Laboratory) under my supervision.

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ARAVINTHAA S (2116230701032)

ABSTRACT

The Campus Food Ordering App is a mobile application developed using Kotlin in Android Studio to simplify and modernize the food ordering process within a college campus. The application replaces the traditional kiosk or scanner-based ordering system by providing a digital platform where students can conveniently browse food shops, view menus, place orders, and track order status in real time using their smartphones.

The system supports multiple food outlets available inside the campus, including Rec Mart, Hut Cafe, Rec Cafe, 6Senses, Dominos, Blackbuck, Coffee Day, Hekka, and Courtyard. The application consists of two main modules: the Student Module and the Shopkeeper Module. Students can register and log in securely using Firebase Authentication, select food items, add them to a cart, place orders, and receive a unique order ID for tracking. Shopkeepers can manage incoming orders, update order status such as Preparing or Ready, and verify orders using the generated order ID or QR code.

Firebase Firestore is used as the backend database for storing user details, shop information, menus, and order data. Real-time database updates ensure instant synchronization between students and shopkeepers, improving communication and reducing waiting time. RecyclerView is used for efficient menu display, while XML layouts are used to create a clean and user-friendly interface.

The proposed system enhances convenience, reduces crowding near food counters, minimizes manual errors, and improves the overall food ordering experience within the campus. This project demonstrates the integration of Android application development with cloud-based backend services to build a practical real-time ordering solution.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
DB	Database
UI	User Interface
XML	Extensible Markup Language
IDE	Integrated Development Environment
SDK	Software Development Kit
QR	Quick Response
Firebase	Google Firebase Cloud Platform
Kotlin	Kotlin Programming Language for Android
RecyclerView	Android UI Component for Efficient List Rendering

CHAPTER 1
INTRODUCTION

1.1 GENERAL

The rapid growth of mobile technology has transformed the way people access services in their daily lives. Food ordering applications have become increasingly popular because they provide convenience, reduce waiting time, and improve customer experience. In many educational institutions, students still depend on traditional food ordering methods such as manual ordering or kiosk-based systems. These methods often create long queues, delays, confusion during peak hours, and difficulties in order management.

The Campus Food Ordering App is developed to provide a smart and efficient digital solution for food ordering within a college campus. The application enables students to browse available food shops, view menu items, add items to the cart, place orders, and track order status directly from their mobile phones. The application also allows shopkeepers to manage incoming orders and update order status in real time.

The project is developed using Kotlin in Android Studio with Firebase Authentication and Firebase Firestore as backend services. The application aims to reduce crowding, minimize manual errors, and improve the overall food ordering experience inside the campus.

1.2 OBJECTIVE

The main objectives of the Campus Food Ordering App are:

1. To develop a mobile application for campus food ordering.

2. To reduce waiting time and long queues in food courts.
3. To provide secure user authentication using Firebase.
4. To allow students to browse multiple food shops easily.
5. To enable online food ordering through smartphones.
6. To provide real-time order status tracking.
7. To help shopkeepers efficiently manage orders.
8. To improve communication between students and food vendors.
9. To generate order IDs and QR codes for order verification.
10. To provide a user-friendly and efficient ordering system.

1.3 EXISTING SYSTEM

The existing food ordering system in many college campuses mainly depends on manual ordering or kiosk/scanner-based systems. Students need to stand in long queues to place orders, especially during lunch and break times. Shopkeepers manually manage orders, which increases the chances of mistakes and delays.

1.3.1 Disadvantages of Existing System

1. Long waiting time during busy hours.
2. Students must stand in queues for ordering.
3. Manual order handling may lead to errors.
4. No proper order tracking system.
5. Difficult to manage multiple food shops efficiently.
6. Poor communication between students and vendors.
7. Time-consuming order verification process.
8. Lack of digital management and automation.

These limitations create the need for a smart mobile-based food ordering system.

1.4 PROPOSED SYSTEM

The proposed system is a mobile application designed to simplify food ordering inside the campus. Students can use the application to order food from multiple campus food shops without physically visiting the counters.

The application provides separate functionalities for students and shopkeepers. Students can register, log in, browse menus, add items to cart, place orders, and track order status. Shopkeepers can manage orders, update order status, and verify orders using QR codes or order IDs.

Firebase Authentication is used for secure login and registration, while Firebase Firestore stores all user, menu, and order data in real time.

1.4.1 Advantages of Proposed System

1. Reduces crowding and waiting time.
2. Provides a smooth digital ordering experience.
3. Real-time order tracking.
4. Easy management of multiple food shops.
5. Secure authentication and cloud storage.
6. Efficient order verification process.
7. User-friendly interface.
8. Faster communication between users and shopkeepers.

CHAPTER 2

LITERATURE REVIEW

2.1 GENERAL

Several mobile food ordering systems have been developed to improve customer convenience and reduce manual work in restaurants and cafeterias. Modern applications use cloud-based databases and real-time synchronization to manage orders effectively.

Research on mobile food ordering systems shows that digital ordering platforms significantly reduce waiting time and improve order accuracy. Many applications use Android technology because of its flexibility, open-source environment, and large user base.

Cloud platforms such as Firebase are commonly used in mobile applications for authentication, database management, and real-time updates. Firebase provides secure and scalable backend support, making it suitable for applications that require real-time synchronization.

Studies on real-time ordering systems also indicate that QR code-based verification improves order security and reduces fraudulent pickups. Real-time order tracking helps users know the preparation status of their food, thereby improving customer satisfaction.

The Campus Food Ordering App combines these modern technologies to create an efficient food ordering solution specifically designed for educational institutions.

CHAPTER 3

SYSTEM DESIGN

3.1 GENERAL

The Campus Food Ordering App follows a client-server architecture where the Android application acts as the client and Firebase acts as the backend server. The application is divided into different modules such as authentication, menu management, cart management, order processing, and order tracking.

The system is designed to provide secure communication between users and the database while ensuring real-time synchronization of order information.

3.1.1 SYSTEM FLOW DIAGRAM

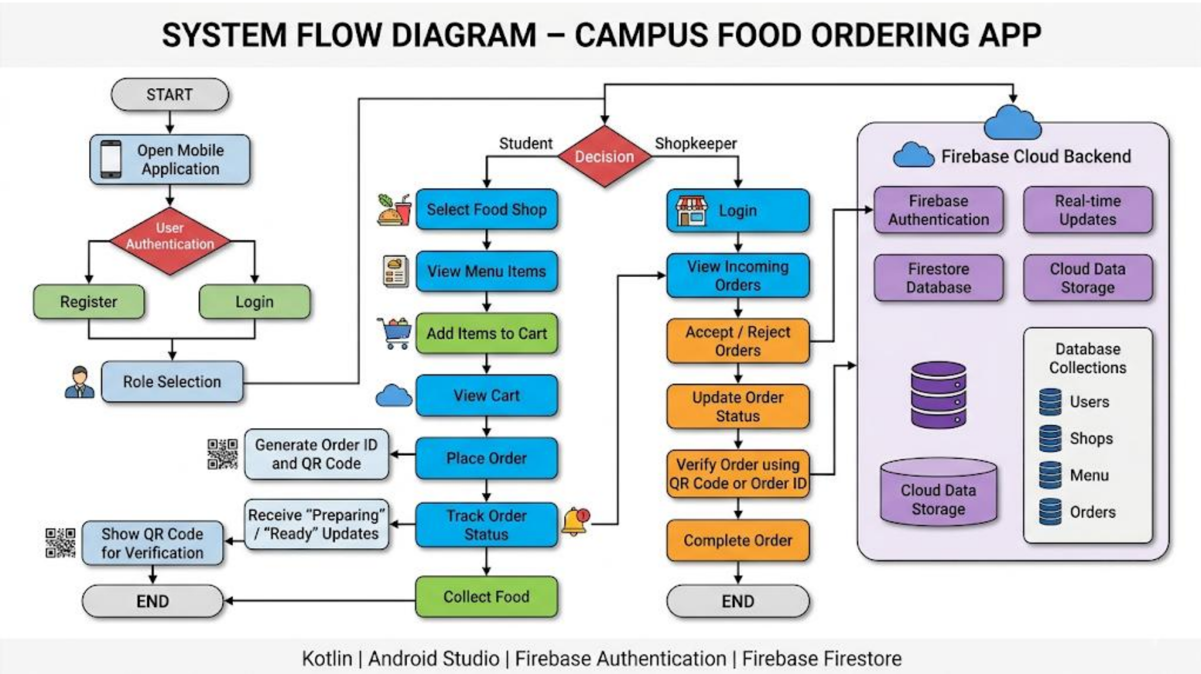

Figure 3.1: System Flow Diagram

The system flow for the Campus Food Ordering App is described below:

1. User opens the application.
2. User registers or logs in using Firebase Authentication.
3. User selects a food shop from the list of available outlets.
4. Menu items for the selected shop are displayed.
5. User adds desired items to the cart.
6. User places the order from the cart screen.
7. Order details are stored in Firebase Firestore with a unique Order ID.
8. Shopkeeper receives order notification and begins preparation.
9. Shopkeeper updates order status (Preparing / Ready).
10. User tracks order status in real time on the tracking screen.
11. QR code or Order ID is used for order verification during pickup.

3.1.2 ARCHITECTURE DIAGRAM

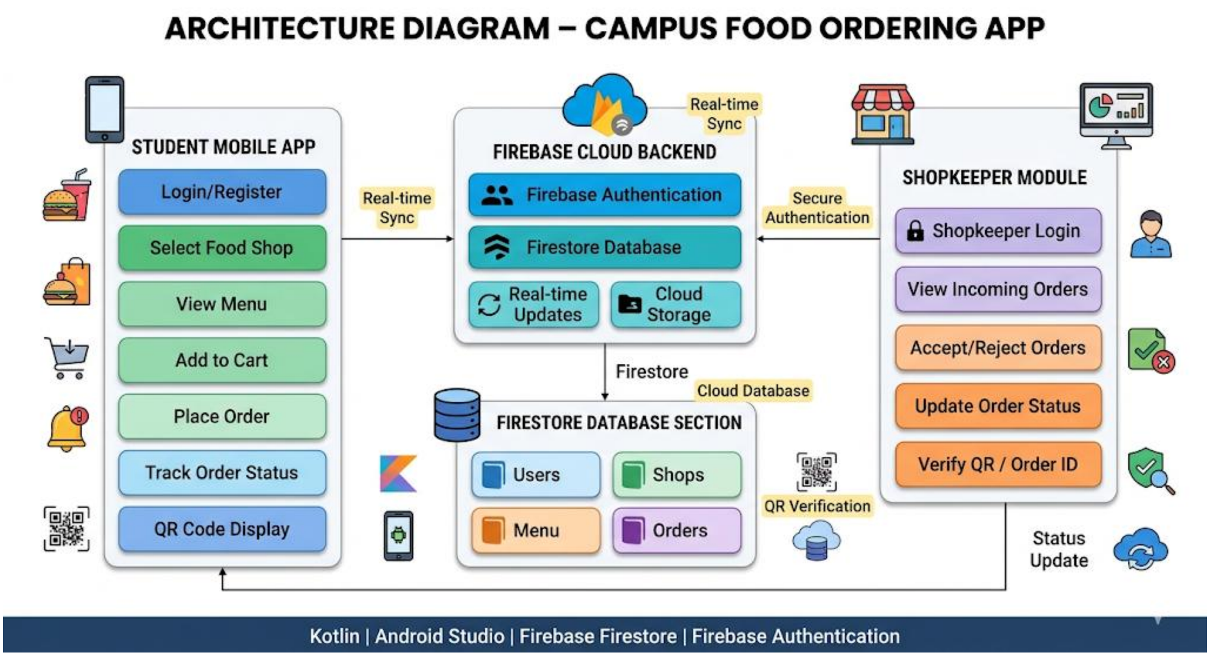

Figure 3.2: System Architecture Diagram

The architecture of the Campus Food Ordering App consists of the following components:

User Module

Allows students to: Login/Register, Select shops, View menu, Add items to cart, Place order, and Track orders.

Shopkeeper Module

Allows shopkeepers to: View incoming orders, Accept or reject orders, Update order status, and Verify orders using QR code.

Firebase Backend

Firebase handles: Authentication, Firestore database management, and Real-time data updates.

Database Collections

- users – Stores user account and role information
- shops – Stores food outlet details
- menu – Stores food items and prices per shop
- orders – Stores order details, status, and user references

3.1.3 USE CASE DIAGRAM

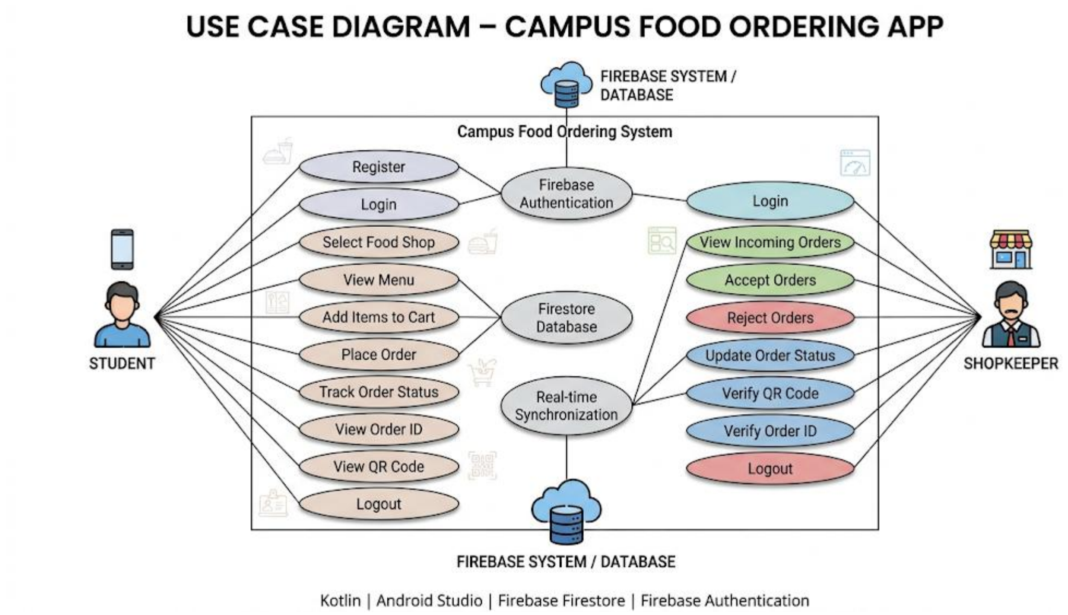

Figure 3.3: Use Case Diagram

Student Use Cases

- Register
- Login
- Select Shop
- View Menu
- Add to Cart
- Place Order
- Track Order

Shopkeeper Use Cases

- Login
- View Orders
- Update Order Status
- Verify Orders using QR Code

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

The project is developed using Android Studio and Kotlin programming language. Firebase services are integrated to provide authentication and cloud database support. The development process follows a structured approach to ensure quality and reliability of the application.

The development process includes the following phases:

11. Requirement analysis – Understanding student and shopkeeper needs.
12. UI design using XML – Designing clean and intuitive user interfaces.
13. Firebase integration – Connecting the application to Firebase Authentication and Firestore.
14. Module development – Building each functional module independently.
15. Order management implementation – Handling order placement and status updates.
16. Real-time synchronization – Ensuring instant data sync between student and shopkeeper modules.
17. QR code integration – Generating and verifying QR codes for order pickup.
18. Application deployment – Testing and releasing the application.

The application uses RecyclerView for displaying lists efficiently and Firebase Firestore for storing and retrieving real-time data.

SYSTEM REQUIREMENTS

Table 4.1: Hardware Requirements

Hardware Component	**Specification**
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Storage	Minimum 10 GB free disk space
Mobile Device	Android Smartphone (Android 6.0 or above)
Network	Internet connection required

Table 4.2: Software Requirements

Software Component	**Specification**
Operating System	Windows 10 / Windows 11
IDE	Android Studio (Latest Version)
Programming Language	Kotlin
Backend Authentication	Firebase Authentication
Cloud Database	Firebase Firestore
UI Design	XML Layouts
SDK	Android SDK

4.1.1 MODULES

Module 1: Login and Registration Module

This module allows users to create accounts and log in securely using Firebase Authentication. Both students and shopkeepers access the application through this module with role-based permissions.

Features:

- Email and password-based authentication
- Secure login using Firebase Authentication
- Role-based access for Student and Shopkeeper

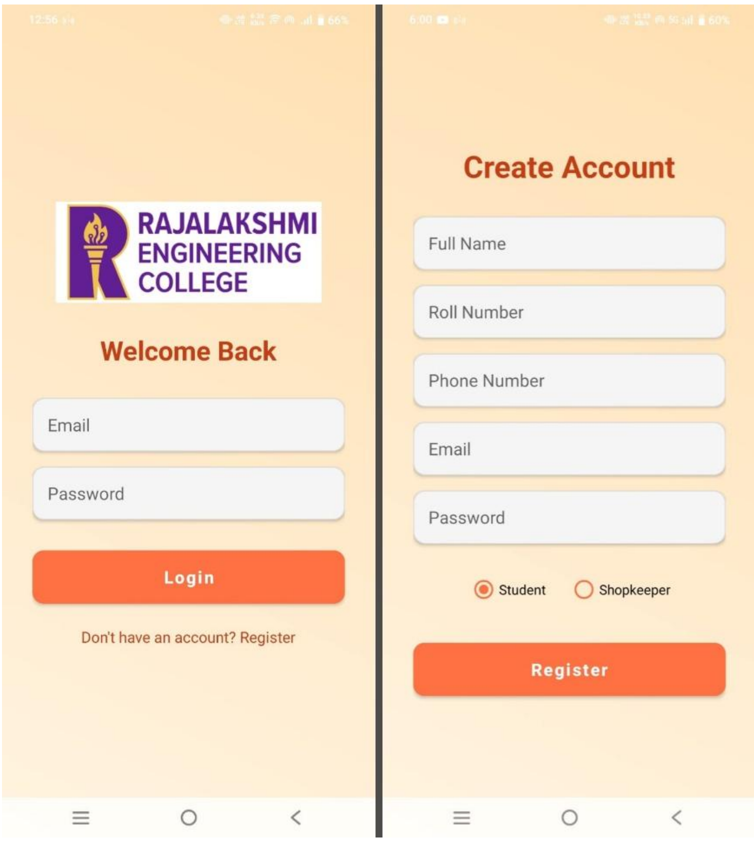

Figure 4.1: Login and Registration Screen

Module 2: Shop Selection Module

This module displays all available campus food shops using RecyclerView. Students can select a shop to view its menu and proceed with ordering.

Available Shops:

- Rec Mart
- Hut Cafe
- Rec Cafe
- 6Senses
- Dominos
- Blackbuck
- Coffee Day
- Courtyard

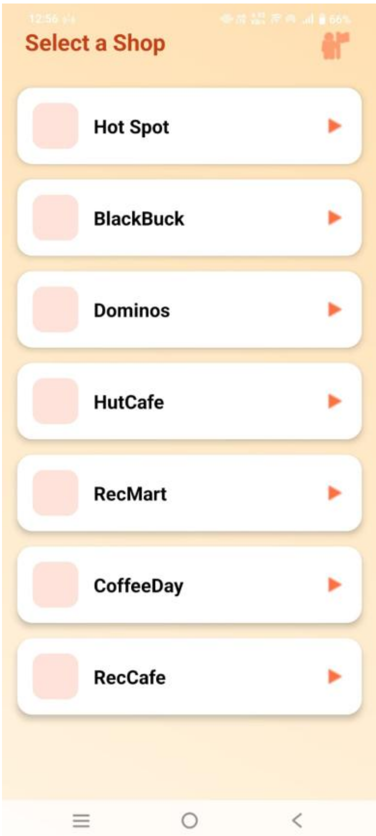

Figure 4.2: Shop Selection Screen

Module 3: Menu Display Module

The Menu Display Module shows all food items and their prices for the selected shop. Students can browse items and add them to their cart.

Features:

- Food item name and description display
- Price display for each item
- Add to cart option for each item

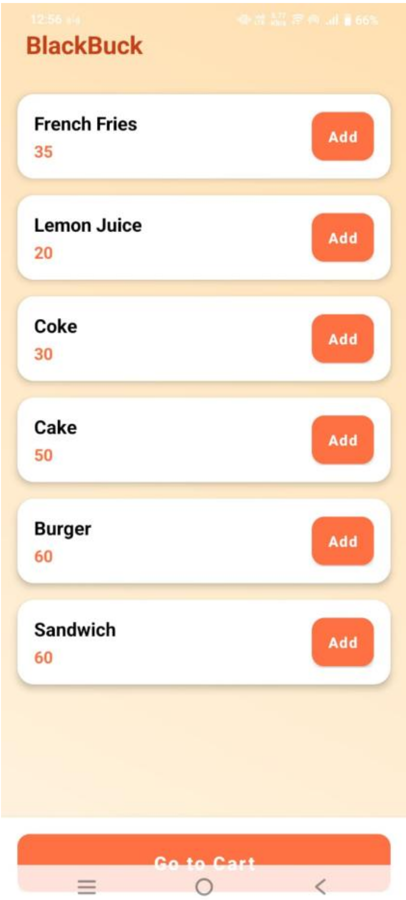

Figure 4.3: Menu Display Screen

Module 4: Cart Module

The Cart Module stores all selected food items before order placement. Students can review their selections, adjust quantities, and proceed to place the order.

Features:

- View all selected items
- Quantity management for each item
- Total price calculation
- Place order functionality

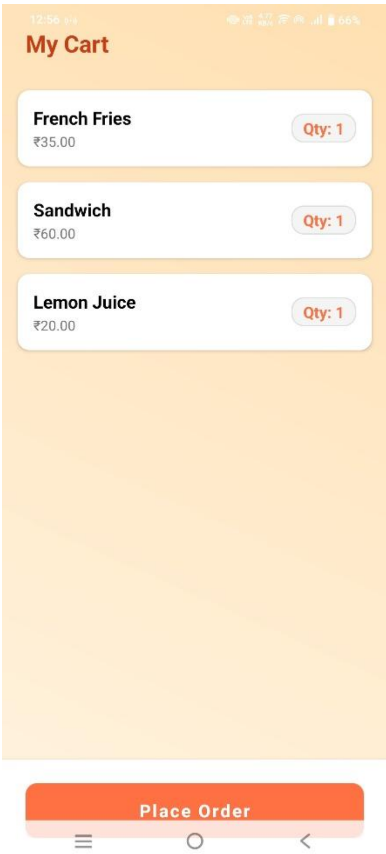

Figure 4.4: Cart Screen

Module 5: Order Placement Module

This module handles order submission and stores order details in Firebase Firestore. A unique Order ID is automatically generated for each order placed.

Features:

- Generate unique Order ID and QR for each order
- Save order details in Firebase Firestore

- Real-time order status updates

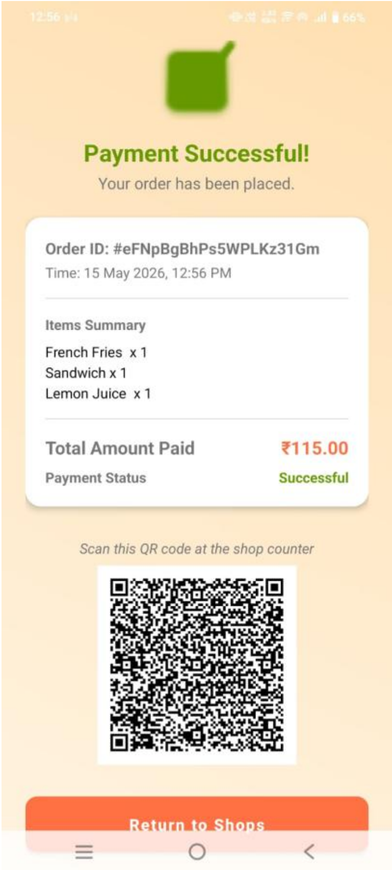

Figure 4.5: Order Placement Screen

Module 6: Order Tracking Module

The Order Tracking Module allows students to monitor the real-time status of their placed orders. The status is updated by the shopkeeper and reflected instantly on the student's screen.

Order Status Values:

- Preparing – Order is being prepared by the shopkeeper
- Ready – Order is ready for pickup

Module 7: QR Code Verification Module

The QR Code Verification Module generates a unique QR code for each order. The shopkeeper scans this QR code during pickup to verify the order and mark it as completed.

Features:

- QR code generated automatically for each placed order
- Shopkeeper scans QR code to verify the order
- Secure and accurate order pickup process

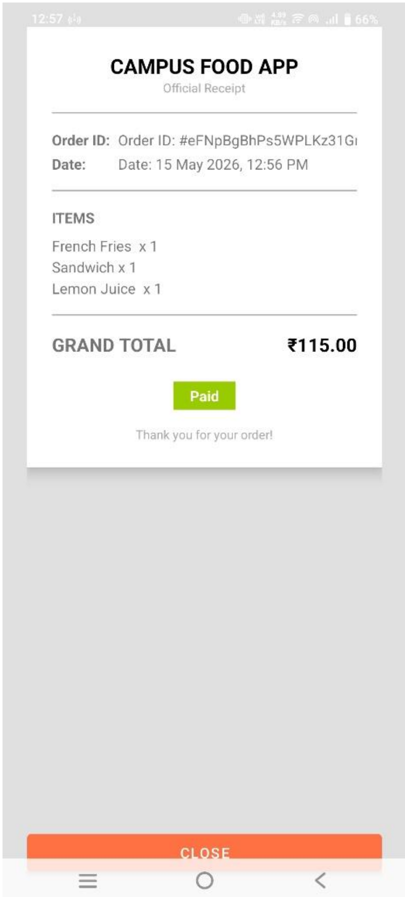

Figure 4.6: QR Code Verification Screen

Module 8: Firebase Integration Module

Firebase services are integrated throughout the application to provide authentication, cloud data storage, and real-time synchronization between the student and shopkeeper modules.

Firebase Services Used:

- Firebase Authentication – Secure user login and registration
- Firebase Firestore – Cloud database for storing all application data
- Real-time synchronization – Instant updates between student and shopkeeper

Firestore Collections:

- users – Stores user account and role information
- shops – Stores food outlet details
- menu – Stores food items and prices per shop
- orders – Stores order details, status, and user references

Table 4.3: users Collection

Field Name	Data Type	Description
userId	String	Unique identifier for each user
email	String	Registered email address of the user
role	String	User role: Student or Shopkeeper

Table 4.4: shops Collection

Field Name	Data Type	Description
shopId	String	Unique identifier for each shop
shopName	String	Name of the food outlet

Table 4.5: menu Collection

Field Name	Data Type	Description
itemName	String	Name of the food item
itemPrice	Number	Price of the food item
shopId	String	Reference to the shop offering this item

Table 4.6: orders Collection

Field Name	Data Type	Description
orderId	String	Unique identifier for each order
userId	String	Reference to the user who placed the order
items	Array	List of ordered food items with quantities
status	String	Current order status: Preparing / Ready / Completed

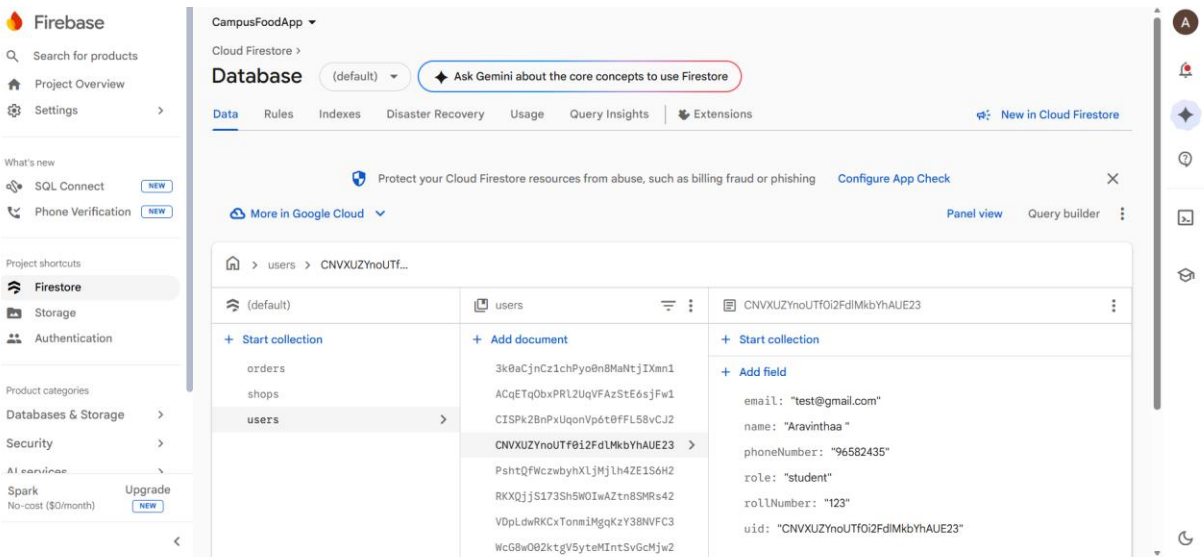

Figure 4.7: Firebase Firestore Database Structure

CHAPTER 5

CONCLUSIONS

5.1 GENERAL

The Campus Food Ordering App successfully provides a smart and efficient solution for managing food ordering within a college campus. The application simplifies the ordering process for students and improves order management for shopkeepers.

The integration of Firebase Authentication and Firebase Firestore ensures secure and real-time handling of user and order data. The application reduces waiting time, minimizes manual errors, and enhances the overall user experience.

The project demonstrates the practical implementation of Android mobile application development using Kotlin and Firebase technologies. The developed system is scalable, user-friendly, and suitable for modern campus environments.

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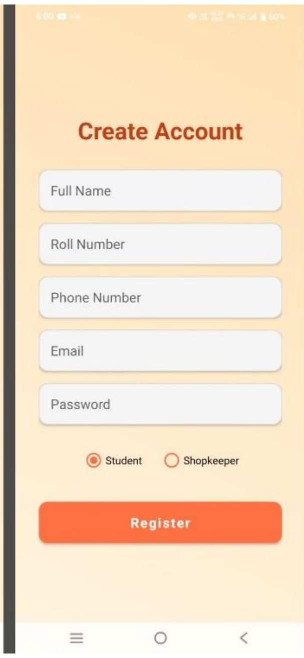
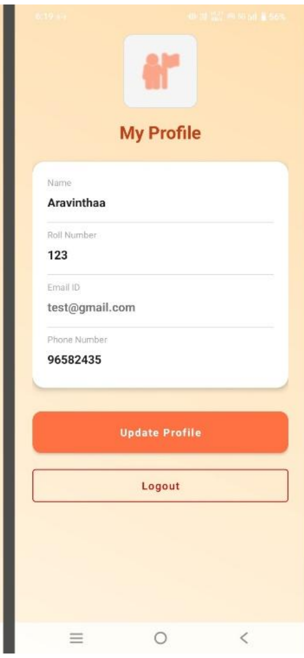
APPENDIX A – SCREENSHOTS

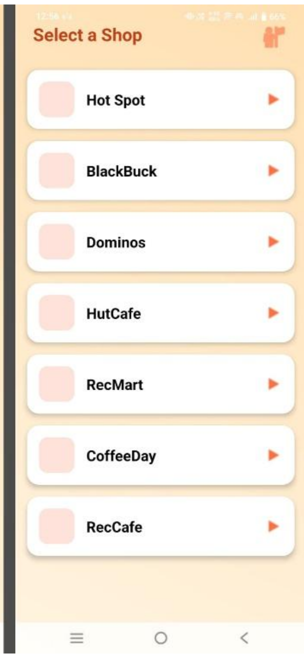
Login Screen



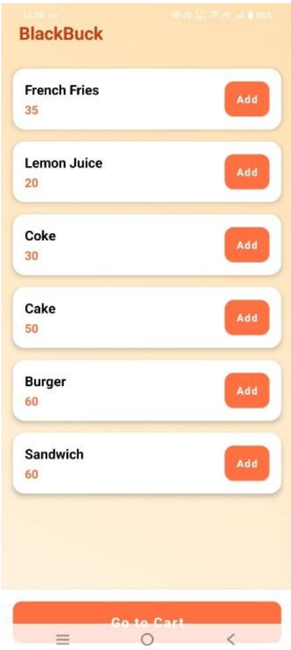
Registration Screen



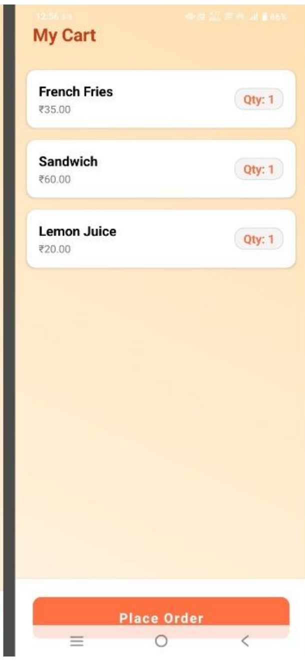
User Profile Page



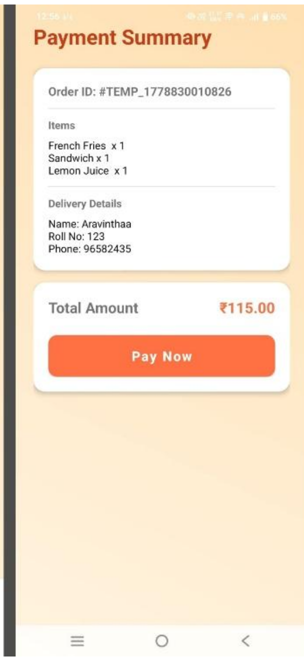
Shop Selection Screen



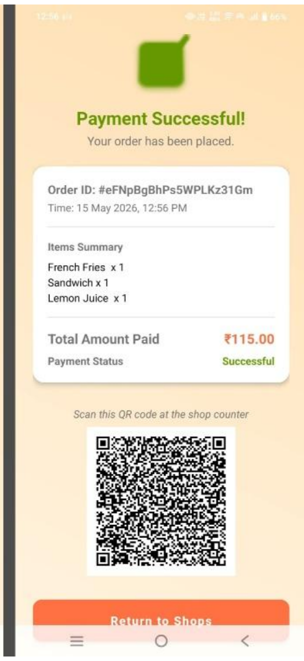
Menu Display Screen



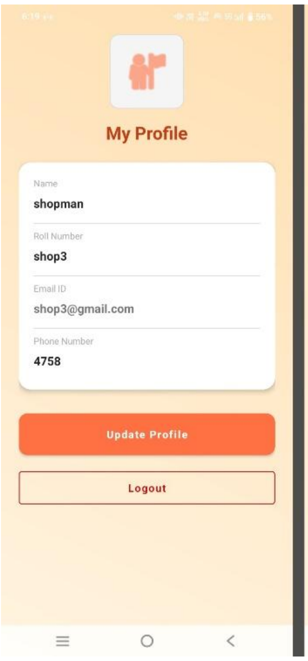
Cart Screen



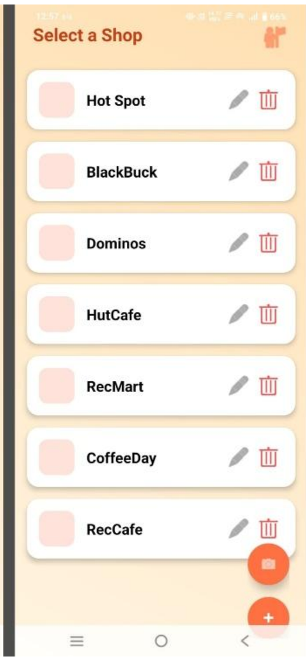
Payment Screen



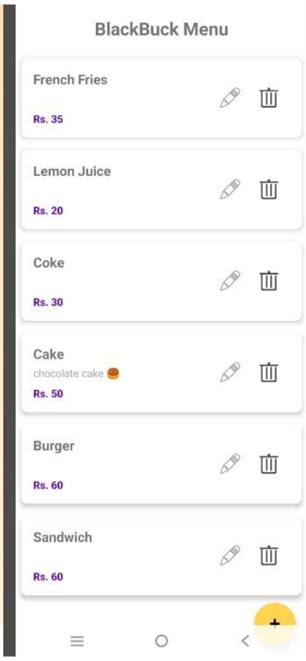
Order Placement Screen



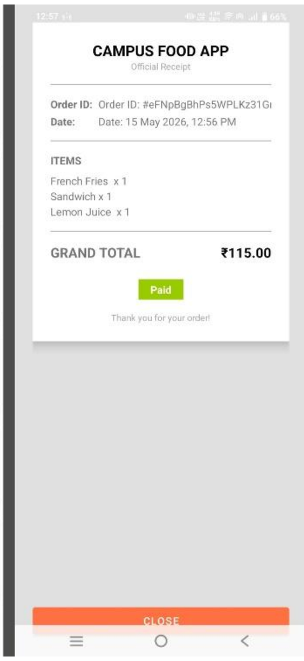
Shopkeeper Profile



Shop Management



Items Management



Order Confirmation [Receipt]

APPENDIX B – SOURCE CODE FILES

XML LAYOUT FILES

1. activity_cart.xml – Cart screen layout
2. activity_login.xml – Login screen layout
3. activity_main.xml – Main activity layout
4. activity_menu.xml – Menu display screen layout
5. activity_order_confirmation.xml – Order confirmation screen layout
6. activity_payment.xml – Payment screen layout
7. activity_profile.xml – User profile screen layout
8. activity_receipt.xml – Receipt generation screen layout
9. activity_register.xml – Registration screen layout
10. activity_shop.xml – Shop selection screen layout
11. activity_shopkeeper_menu.xml – Shopkeeper menu management screen layout
12. dialog_add_menu_item.xml – Dialog layout for adding menu items
13. item_cart.xml – Cart item RecyclerView layout
14. item_menu.xml – Menu item RecyclerView layout
15. item_menu_shopkeeper.xml – Shopkeeper menu item layout
16. item_shop.xml – Shop item RecyclerView layout

ACTIVITY FILES

1. MainActivity.kt – Main entry point of the application
2. LoginActivity.kt – Handles user login functionality
3. RegisterActivity.kt – Handles user registration functionality
4. ShopActivity.kt – Displays available food shops
5. MenuActivity.kt – Displays menu items of selected shop
6. CartActivity.kt – Handles cart operations
7. OrderConfirmationActivity.kt – Displays order confirmation details

8. PaymentActivity.kt – Handles payment-related functionality
9. ProfileActivity.kt – Displays user profile information
10. ReceiptActivity.kt – Generates and displays order receipt
11. ShopkeeperMenuActivity.kt – Allows shopkeeper to manage menu items
12. CaptureActivityPortrait.kt – Handles QR code scanning in portrait mode

ADAPTER FILES

1. CartAdapter.kt – RecyclerView adapter for cart items
2. MenuAdapter.kt – RecyclerView adapter for menu items
3. ShopAdapter.kt – RecyclerView adapter for shop list
4. ShopkeeperMenuAdapter.kt – RecyclerView adapter for shopkeeper menu management

MODEL FILES

1. MenuItem.kt – Data model for menu items
2. Order.kt – Data model for orders
3. Shop.kt – Data model for food shops
4. User.kt – Data model for user details

FIREBASE FILES

1. Firebase Authentication Integration – Handles user authentication
2. Firebase Firestore Integration – Handles cloud database operations
3. Real-time Database Synchronization – Updates order status in real time

UTILITY FILES

1. Utility Classes – Helper functions and common reusable methods